Algorithmic Trading

Transaction Cost Analysis (TCA)

Post-Trade and Pre-Trade Formulas

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<u>Algo-TCA Suite Notation:</u>

Notation:

$S =$ Order Shares

$X =$ Executed Shares

$R =$ Residual (Unexecuted) Shares

$P_d =$ Decision Price – Price at the time of the investment decision (mid-point of bid-ask spread)

$P_0 =$ Arrival Price – Price at the time the order was entered into the market (mid-point of bid-ask spread)

$P_n =$ Price at end of trading. Often as the closing price on the day.

$P_{avg} =$ Average execution price of the order.

$P_i =$ Price of the ith trade.

$V_i =$ Volume of the ith trade (e.g., shares traded in the ith trade).

$VWAP =$ Volume weighted average price (includes all market volume).

$Open =$ Opening price on the day (official open from the exchange where the stock is listed).

$Close =$ Closing price on the day (official close from the exchange where the stock is listed).

$P_b =$ Price benchmark defined by the analyst. The price benchmark can be Open, Close, VWAP, Arrival, etc.

$Open\ T+1 =$ the next day's opening price. E.g., the opening price on the day after completion of the trade or cancellation of the order.

$Close\ T+1 =$ the next day's closing price. E.g., the closing price on the day after completion of the trade or cancellation of the order.

$Side =$ order side, +1 if buy order, -1 if sell order.

$Fixed =$ fixed costs of the order. Includes broker commission and SEC taxes (if applicable).

- $Commission =$ per share cost for all executed shares.
- $SEC\ Tax =$ per share cost for sell orders only.

$Delay\ Cost =$ the cost due to the price movement in the stock from the time of the investment decision and the time the order was released into the market. This cost is attributable to the asset management firm and/or market price movement. It is often caused by the asset management firm having to determine which broker, algorithm, and/or algorithmic parameter for the order.

$Execution\ Cost$ = the cost incurred during trading. Execution cost is also referred to as arrival cost and trading cost. Measured as the difference between the average execution price and the price at the time the order was entered into the market (arrival price). The execution cost of the order is attributable to the broker and/or algorithmic.

$Opportunity\ Cost$ = the missed profit opportunity caused by not being able to execute the entire order in the market. Opportunity cost is attributable to the residual (unexecuted) shares of the order.

$Trade\ Quantity$ = trade quantity, expressed as Size (%ADV), Shares, Trade Value.

$Size$ = order size (%ADV) calculated as Order Shares divided by Average Daily Volume (ADV) and expressed as a decimal. Also referred to as %ADV.

$\%ADV$ = order size expressed as a percentage of ADV, %ADV. Calculated as Order Shares divided by Average Daily Volume (ADV) and expressed as a decimal. Also referred to as size or order size.

ADV = average daily volume of the stock.

$Volatility$ = annualized log return volatility expressed as a decimal.

POV = percentage of volume trading rate, expressed as a decimal.

$AlphaBp$ = price movement over the day expressed in basis points.

$Trade\ Time$ = trade time expressed as a percentage of the day as a decimal in volume time. Also denoted as Time, “t,” or “T.”

Bid = the broker’s capital commitment or principal bid for an order.

a_1, a_2, a_3, a_4, b_1 = market impact model parameters.

λ = investor’s level of risk aversion

Post-Trade TCA Formulas:

Implementation Shortfall

$$Paper\ Return = Side \cdot S \cdot (P_n - P_d)$$

$$Portfolio\ Return = Side \cdot X \cdot \left(P_n - P_{avg}\right) - Fixed$$

$$IS = Paper\ Return - Actual\ Return$$

Implementation Shortfall – Cost Decomposition

$$IS_{\$} = Side \cdot \left\{ \underbrace{S \cdot (P_0 - P_d)}_{Delay\ Cost} + \underbrace{X \cdot \left(P_{avg} - P_0\right)}_{Execution\ Cost} + \underbrace{R \cdot (P_n - P_0)}_{Opportunity\ Cost} \right\} + Fixed$$

Arrival Cost

The arrival cost of the order. Measured as the difference between the average execution price P_{avg} and the order arrival price P_0. This is a measure of the trading cost attributable to the broker and/or algorithms. It is calculated as follows:

$$Arrival\ Cost(bp) = Side \cdot \frac{P_{Avg} - P_0}{P_0} \cdot 10^4$$

VWAP Slippage

A measure of trading performance compared to the VWAP price. The VWAP price is an indication of the fair or market price.

$$VWAP\ Slippage(bp) = Side \cdot \frac{P_{Avg} - VWAP}{VWAP} \cdot 10^4$$

Benchmark Cost

Benchmark Cost for a specified benchmark price P_b is calculated as follows:

$$Benchmark\ Cost(bp) = Side \cdot \frac{P_{Avg} - P_b}{P_b} \cdot 10^4$$

Relative Performance Measure (RPM):

$$\text{RPM}(Buy) = \frac{\left(Volume\ at\ Price > P_{Avg}\right) + \left(\frac{1}{2} \cdot Volume\ at\ Price = P_{Avg}\right)}{Total\ Volume}$$

$$\text{RPM}(Sell) = \frac{\left(Volume\ at\ Price < P_{Avg}\right) + \left(\frac{1}{2} \cdot Volume\ at\ Price = P_{Avg}\right)}{Total\ Volume}$$

Broker Value-Add

$$VA = E[Cost] - Arrival\ Cost$$

$$ZVA = \frac{E[Cost] - Arrival\ Cost}{TR} = \frac{Value\ Add}{TR}$$

Pre-Trade Formulas:

MI Parameters:

Market Impact parameters consist of the following:

a_1 = Multiplier

a_2 = Size Shape

a_3 = Volatility Shape

a_4 = POV Shape

b_1 = Pct Temporary Impact $(0 \leq b_1 \leq 1)$

Market Impact:

$$I^* = a_1 \cdot Size^{a2} \cdot \sigma^{a3}$$

$$MI = I^* \cdot \left(b_1 \cdot POV^{a4} + (1 - b_1)\right)$$

Timing Risk:

$$TR = \sigma \cdot \sqrt{\frac{1}{3} \cdot \frac{1}{250} \cdot Size \cdot \frac{(1 - POV)}{POV}} \cdot 10^4 bp$$

$$TR = \sigma \cdot \sqrt{\frac{1}{3} \cdot \frac{1}{250} \cdot Trade\ Time} \cdot 10^4 bp$$

Price Appreciation:

$$PA = Side \cdot \frac{1}{2} \cdot AlphaBp \cdot Size \cdot \frac{(1 - POV)}{POV}$$

$$PA = Side \cdot \frac{1}{2} \cdot AlphaBp \cdot Trade\ Time$$

Expected Cost & Risk:

$$E(Cost) = MI + PA$$

$$E(Risk) = TR$$

Expected Price:

$$P_{avg} = P_0 \cdot \left(1 + Side \cdot \frac{MI + PA}{10000}\right)$$
$$= P_0 \cdot \left(1 + Side \cdot \frac{E(Cost)}{10000}\right)$$

Conversions:

POV Rate to Trade Time:

$$Trade\ Time = Size \cdot \frac{(1 - POV)}{POV}$$

Trade Time to POV Rate:

$$POV = \frac{Size}{Size + Time}$$

Risk Analysis

Confidence Intervals:

$$P_{avg} = P_0 \cdot \left(1 + Side \cdot \frac{MI + PA}{10000}\right)$$

$$SE = P_0 \cdot \left(1 + \frac{TR}{10000}\right)$$

$$LB(\alpha) = P_{avg} - \left|Z_{\alpha/2}\right| \cdot SE$$

$$UB(\alpha) = P_{avg} + \left|Z_{\alpha/2}\right| \cdot SE$$

Value-at-Risk (VaR)

$$P_{avg} = P_0 \cdot \left(1 + Side \cdot \frac{MI + PA}{10000}\right)$$

$$SE = P_0 \cdot \frac{TR}{10000}$$

$$VaR(\alpha) = P_{avg} - |Z_{\alpha}| \cdot SE$$

Confidence Interval (two-tail, alpha/2)							
C.I.	99.8%	99.0%	98.0%	95.0%	90.0%	80.0%	50.0%
alpha	0.001	0.005	0.010	0.025	0.050	0.100	0.250
Z	-3.090	-2.576	-2.326	-1.960	-1.645	-1.282	-0.674

Value-at-Risk (one-tail, alpha)							
VaR	99.9%	99.5%	99.0%	99.0%	95.0%	90.0%	75.0%
alpha	0.001	0.005	0.010	0.010	0.050	0.100	0.250
Z	-3.090	-2.576	-2.326	-2.326	-1.645	-1.282	-0.674

Optimization Formulas:

Traders Dilemma

$$Min: \quad MI + \lambda \cdot TR$$

Min Trade Cost

$$Min: \quad MI + PA$$

Price Improvement

$$Max: \quad \frac{Bid - MI}{TR}$$

$$\rightarrow \quad Min: \; -1 \cdot \frac{Bid - MI}{TR}$$

$$\rightarrow Min: \; \frac{MI - Bid}{TR}$$

Price Formulas

VWAP Price

The volume-weighted average price for all trades over a specified period of time (e.g., 1-day, etc.). It is calculated as follows:

$$VWAP\ Price = \frac{\sum_{j=1}^{n} P_j \cdot V_j}{\sum_{j=1}^{n} V_j}$$

P_j = market price for the jth trade.

V_j = market volume of the jth trade.

Average Price

Investor's average execution price:

$$Avg\ Price = \frac{\sum_{i=1}^{m} p_i \cdot s_i}{\sum_{i=1}^{m} s_i}$$

p_i = investor's execution price for their ith trade.

s_i = investor's executed number of shares for their ith trade.